FICTIONAL TEST PACKET. Not a real application.

Applicant's Name: Marcus Hale

STATEMENT OF PURPOSE

I am applying to the MSAI program to study optimization for models that do not fit on one machine. I maintain an open-source library, SmallBatch, that implements a few stochastic solvers for people who want a readable PyTorch training loop rather than a framework. About forty GitHub accounts have starred it, and I have merged patches from six outside contributors. The users are students. I do not want to overstate that.

The mathematics degree is the other half. I have had linear algebra, real analysis, probability, numerical methods, and a graduate-level convex optimization course as an undergraduate. What I do not have is a lab appointment or a paper. The library is the technical artifact. Northwestern is the program I want because the MSAI curriculum pairs the optimization coursework with systems, which is the gap between my solvers and a model someone else can train.

I hope to spend the year on distributed training and a project that turns one of the SmallBatch solvers into something a course staff could assign. I am applying for that sequence, not for a general credential.

EDUCATION

Undergraduate institution: University of Wisconsin-Madison
Degree: B.S. Mathematics and Computer Sciences, 2025
Cumulative GPA: 3.74

RESEARCH

No formal research appointment. Independent work on SmallBatch, described under projects, from June 2024 to the present.

PROJECTS

SmallBatch, an open-source PyTorch training library. I wrote the solvers, the tests, and the documentation. Six outside contributors have merged patches. The issue tracker shows students using it for class projects. Repository: a public GitHub project under my name, started June 2024.

WORK EXPERIENCE

Tutor, Madison Mathematics Learning Center, September 2023 to May 2025, eight hours a week. I ran drop-in hours for linear algebra.

HONORS AND AWARDS

Van Vleck scholarship for mathematics, 2024. Dean's list, four semesters.

LETTERS OF RECOMMENDATION

Letter 1. Professor Ruth Adler, University of Wisconsin. "Marcus took my convex optimization course and

then built a library whose solvers match the algorithms in the course. I have read the documentation. It is accurate. Marcus is ready for graduate coursework in optimization, and I recommend him strongly."

Letter 2. Professor Ian Brooks, University of Wisconsin. "I supervised Marcus only as an instructor in numerical methods, not as a research advisor. Within that scope, the work was excellent. He is the student other students asked for by name in the tutoring center. I recommend him for MSAI."